Algebraic Topology

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1 Introduction

1.1 Recap of IB Topological Spaces

We use algebra to turn topological questions to algebraic questions. This results in a loss of information but lets us prove some results.

Example.

$$\mathbb{R}^n \cong \mathbb{R}^m \implies n = m$$

This is a statement which is very hard to prove otherwise. Likewise we can distinguish topological surfaces with a different number of 'holes' in their surface.

For this section all functions are continuous until mentioned otherwise.

Definition. (Map) A *map* is a continuous function.

Lemma. (Gluing lemma) Let $f : X \to Y$ be a function and $C, K \subset X$ be closed subsets with $C \cup K = X$. Then f is continuous if and only if the restriction of f on C and K is continuous.

Proof. We have that f is continuous if and only if for all closed sets $D \subset Y$ the preimage of D is closed. Since $f^{-1}(D) = (f^{-1}(D) \cap C) \cup (f^{-1}(D) \cap K)$, this holds if and only if $\forall D \subset Y$ closed we have that $f^{-1}(D) \cap C$ and $f^{-1}(D) \cap K$ are closed. Since C and K are closed in X, a subset of C (resp. K) is closed in C (resp. K) if and only if it is closed in X. Hence this holds if and only if $(f|_C)^{-1}(D), (f|_K)^{-1}(D)$ are closed in C and K respectively. Hence this holds if and only if the restriction of f on C and K is continuous. □

Lemma. (Lebesgue number lemma) Let (X, d) be a compact metric space. For any open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in I}$ of X there exists a $\delta > 0$ such that for all $x \in X$, $B_\delta(x) \subset U_\alpha$ for some $\alpha \in I$.

Proof. Suppose no such δ existed. Then for all $n > 0$ there is some $x_n \in X$ such that $B_{1/n}(x_n) \not\subset U_\alpha$ for all $\alpha \in I$. The sequence $\{x_n\}$ admits a convergent subsequence due to compactness, so *wlog* replace x_n with its convergent subsequence, and take $x_n \to y$. Then take $\alpha \in I$ such that $y \in U_\alpha$. Then since U_α is open there exists a $r > 0$ such that $B_r(y) \subset U_\alpha$. Take $N > 0$ such that

$$\begin{cases} \frac{1}{N} < \frac{r}{2} \\ d(x_N, y) < \frac{r}{2} \end{cases}$$

which implies that $B_{1/N}(x_N) \subset B_r(y) \subset U_\alpha$ which is a contradiction. □

1.2 Cell complexes

Definition. (Cell attachment) Given a space X and a map $f : S^{n-1} \to X$. The space $X \cup_f D^n := X \sqcup D^n / \sim$ where $x \sim f(x), \forall x \in S^{n-1} \subset D^n$ is the result of attaching an n-cell to X along f.
The image of f is called the *attaching sphere*.

Definition. (Cell complex) A *finite-dimensional cell complex* is obtained by:

(i) Start with a set X^0 with the discrete topology.

(ii) Inductively once X^{n-1} is defined we form X^n by attaching a collection of n-cells along some collection of attachment maps

$$\{f_\alpha : S^{n-1} \to X^{n-1}\}_{\alpha \in I}.$$

i.e. that $X^n = \left(X^{n-1} \sqcup \bigsqcup_{\alpha \in I} D^n_\alpha\right) / \sim$ where we define $x \sim f_\alpha(x)$ for $x \in S^{n-1}_\alpha = \partial D^n_\alpha$.

(iii) $X = X^k$ for some $k \geq 0$. The minimal such k is called the dimension of X.

Remark. In general we can make sense of infinite dimensional cell complexes. Suppose that we have

$$X^0 \subset X^1 \subset \cdots$$

such that

(i) X^0 is discrete,

(ii) X^n is formed by attaching n-cells to X^{n-1}.

Define

$$X = \bigcup_{n \geq 0} X^n$$

with the topology where $U \subset X$ is open $\iff U \cap X^n$ is open in X^n for all n. We say that X is a CW-complex and that X^n is the n-skeleton of X.

2 Homotopy and the fundamental group

2.1 Homotopy

Definition. We denote by I the closed interval $[0, 1]$.

Definition. (Homotopy) Let $f, g : X \to Y$. Then we say that a map $H : X \times I \to Y$ is a *homotopy from f to g* if $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$. If such an H exists we say that f is *homotopic* to g and $f \simeq g$.

If $A \subset X$ and $f|_A = g|_A$ then a homotopy H from f to g is relative to A (i.e. a homotopy rel A) if $H(a, t) = f(a) = g(a)$ for all $a \in A$ and $t \in I$. If such an H exists we write $f \simeq_A g$.

Proposition. The relation homotopic rel A defines an equivalence relation.

Proof.

(i) $f \simeq_A f$ trivially by $H(x, t) = f(x)$.

(ii) $f \simeq_A g \implies g \simeq_A f$, by replacing t with $1 - t$.

(iii) Suppose that $f \simeq_A g, g \simeq_A h$. Then let H_1 be the homotopy rel A from f to g and H_2 for g to h. Then define

$$H(x,t) = \begin{cases} H_1(x,2t) & 0 \le t \le \frac{1}{2} \\ H_2(x,2t-1) & \frac{1}{2} \le t \le 1 \end{cases}.$$

Since $H_1(x,1) = g(x) = H_2(x,0)$, H is well-defined, and by the gluing lemma (applied to the closed sets $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$) H is continuous. Also $H(a,t) = f(a)$ for all $a \in A$, so H is a homotopy rel A from f to h and therefore $f \simeq_A h$. □

Definition. (Homotopy equivalence) Given a map $f : X \to Y$ we say that f is a *homotopy equivalence* if there exists a $g : Y \to X$ such that $f \circ g \simeq \mathrm{id}_Y$ and $g \circ f \simeq \mathrm{id}_X$. We call g a *homotopy inverse* of f. If there exists a homotopy equivalence $X \to Y$ then X is *homotopy equivalent* to Y and we write that $X \simeq Y$.

Definition. (Contractible) A space X is *contractible* if X is homotopy equivalent to the one-point space $\{*\}$.

Lemma. Let $f_0, f_1 : X \to Y$ be homotopic maps and $g_0, g_1 : Y \to Z$ also be homotopic maps. Then

$$g_0 \circ f_0 \simeq g_1 \circ f_1.$$

Proof. We want to show that $g_0 \circ f_0 \simeq g_0 \circ f_1 \simeq g_1 \circ f_1$.

Choose homotopies H from f_0 to f_1 and H' from g_0 to g_1. Then $g_0 \circ H$ is a homotopy from $g_0 f_0$ to $g_0 f_1$. We also have that $H' \circ (f_1 \times \mathrm{id}_I)$ is a homotopy from $g_0 f_1$ to $g_1 f_1$ hence we're done by transitivity. □

Proposition. Homotopy equivalence is an equivalence relation.

Proof.

(i) $X \simeq X$ is obvious from the identity map.

(ii) $X \simeq Y$ then we have $Y \simeq X$ just by switching the roles of f and g.

(iii) $X \simeq Y$ and $Y \simeq Z$. Then suppose f, g are the homotopy equivalence and its inverse for X and Y, and f', g' for Y and Z. Consider the map $f'f$ from X to Z and its candidate homotopy inverse gg'. By the lemma above,

$$gg'f'f \simeq g\mathrm{id}_Y f = gf \simeq \mathrm{id}_X$$
$$f'fgg' \simeq f'\mathrm{id}_Y g' = f'g' \simeq \mathrm{id}_Z$$

Therefore $f'f$ is a homotopy equivalence and $X \simeq Z$. □

Example. Consider the inclusion

$$\tau : S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\}$$

with homotopy inverse $r : \mathbb{R}^2 \setminus \{0\} \to S^1$ given by projecting each vector onto its unit vector, $r(x) = x/\|x\|$. Since vector normalisation doesn't do anything to unit vectors one of the directions of composition is obvious: $r \circ \tau = \mathrm{id}_{S^1}$. For the other direction,

$$\mathbb{R}^2 \setminus \{0\} \to S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\}$$

we're given the homotopy

$$\begin{aligned} H : (\mathbb{R}^2 \setminus \{0\}) \times I &\to \mathbb{R}^2 \setminus \{0\} \\ (x, t) &\mapsto \frac{x}{t + (1-t)\|x\|} \end{aligned}$$

which is well-defined since the denominator is strictly positive, and is a homotopy from $\tau \circ r$ to $\mathrm{id}_{\mathbb{R}^2 \setminus \{0\}}$. Therefore we have that $S^1 \simeq \mathbb{R}^2 \setminus \{0\}$. Similarly we have that $S^{n-1} \simeq \mathbb{R}^n \setminus \{0\}$.

Example. We have that $\mathbb{R}^n$ is contractible. Let $f : \mathbb{R}^n \to \{0\}$ be the constant map and $g : \{0\} \hookrightarrow \mathbb{R}^n$ be the inclusion. Then

$$f \circ g = \mathrm{id}_{\{0\}}$$

so one direction is trivial. For the other direction, $g \circ f : \mathbb{R}^n \to \mathbb{R}^n$ is the constant map $x \mapsto 0$. Define the straight-line homotopy

$$\begin{aligned} H : \mathbb{R}^n \times I &\to \mathbb{R}^n \\ (x, t) &\mapsto tx. \end{aligned}$$

This is continuous since scalar multiplication $\mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ is continuous. We have $H(x, 0) = 0 = (g \circ f)(x)$ and $H(x, 1) = x = \mathrm{id}_{\mathbb{R}^n}(x)$, so H is a homotopy from $g \circ f$ to $\mathrm{id}_{\mathbb{R}^n}$. Therefore $g \circ f \simeq \mathrm{id}_{\mathbb{R}^n}$, so f is a homotopy equivalence with homotopy inverse g. Hence $\mathbb{R}^n \simeq \{0\}$, i.e. $\mathbb{R}^n$ is contractible.

The same argument works for any $X \subset \mathbb{R}^n$ which is star-shaped about some point x_0 (e.g. any convex set), using $H(x, t) = (1 - t)x_0 + tx$.

Definition. (Deformation Retraction) Let $i : A \hookrightarrow X$ be the inclusion of a subspace. We say that A is a *deformation retract* of X if there is a map $r : X \to A$ such that $r \mid_A = \mathrm{id}_A$ and $i \circ r \simeq \mathrm{id}_X$. If the homotopy is relative to A, then A is a strong deformation retract of X.

Remark. This implies that $X \simeq A$ by the inclusion map from A to X and r from X to A. This is a strictly stronger condition.

Definition. (Retraction) A *retraction* of X to A is a map $r : X \to A$ such that $r \mid_A = \mathrm{id}$.

Remark. A retraction need not be a homotopy equivalence.

Example. X is a space and $x \in X$. Then X retracts to $\{x\}$. But $X \simeq \{x\} \iff X$ is contractible

2.2 Path

Definition. (Path) For a space X and points $x_0, x_1 \in X$, a *path* from x_0 to x_1 is a map $\gamma : I \to X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$.

We write that $\gamma : x_0 \rightsquigarrow x_1$, and if $x_0 = x_1$ we say that γ is a loop based at x_0.

If we have $\gamma : x_0 \rightsquigarrow x_1$ and $\gamma' : x_1 \rightsquigarrow x_2$ then the concatenation of $\gamma \circ \gamma'$ is defined by using the gluing lemma and going in double time which goes from x_0 to x_2.

We also define the inverse path of $\gamma : x_0 \rightsquigarrow x_1$ as $\gamma^{-1}(t) = \gamma(1-t) : x_1 \rightsquigarrow x_0$.

Finally we define the constant path $c_{x_0}(t) = x_0 : x_0 \rightsquigarrow x_0$.

If we define a relation $\sim$ on X by $x \sim y \iff$ there is a path from x to y, these definitions show that $\sim$ is an equivalence relation.

Definition. (Path components) The equivalence classes of $\sim$ are called *path components* of X. The set of all path components of X is denoted by $\pi_0(X)$. If $|\pi_0(X)| = 1$ then we say that X is *path-connected*.

Remark. X is path-connected $\iff$ $\forall x, y \in X$ there exists a path $x \rightsquigarrow y$.

Proposition. Given a map $f : X \to Y$ there is a well-defined associated function

$$\pi_0(f) : \pi_0(X) \to \pi_0(Y)$$

given by $\pi_0(f)[x] = [f(x)]$. Futhermore,

(i) If $f \simeq g$, then $\pi_0(f) = \pi_0(g)$;

(ii) For maps $A \xrightarrow{h} B \xrightarrow{k} C$ we have that

$$\pi_0(k \circ h) = \pi_0(k) \circ \pi_0(h).$$

(iii) $\pi_0(\mathrm{id}_X) = \mathrm{id}_{\pi_0(X)}$.

Proof. Choose $x, x' \in X$ such that $[x] = [x']$. Then there exists a path γ from x to x' in X. Now $f \circ \gamma$ is a path from $f(x)$ to $f(x')$ hence they are in the same path-component so $\pi_0(f)$ is well-defined.

For (i), let $H : X \times I \to Y$ be a homotopy from f to g. Choose any $x \in X$. The map $H(x, \cdot)$ is a path from $f(x)$ to $g(x)$ so $[f(x)] = [g(x)]$ and hence $\pi_0(f) = \pi_0(g)$. (ii) and (iii) are obvious. □

Corollary. If $f : X \to Y$ is a homotopy equivalence then $\pi_0(f)$ is bijective.

Proof. Let $g : Y \to X$ be the homotopy inverse of f so $fg \simeq \mathrm{id}_Y$ hence $\pi_0(f)\pi_0(g) = \pi_0(\mathrm{id}_Y) = \mathrm{id}_{\pi_0(Y)}$. Similary $gf \simeq \mathrm{id}_X$ gives the other direction. Hence $\pi_0(f)$ is a bijection. □

Example. Consider the space $\{0, 1\}$ equipped with the discrete topology. Let's prove it is not contractible. If it were, then we'd have $\pi_0(\{0,1\}) \simeq \pi_0(*)$ (the one point space). This is false because $\{0, 1\}$ has two path-components.

Example. $\mathbb{R}^2$ does not retract to $A = \{(-1,0),(1,0)\}$. Suppose that $r : \mathbb{R}^2 \to A$ is a retraction. Then the composition

$$\pi_0(A) \overset{\pi_0(\tau)}{\to} \pi_0(\mathbb{R}^2) \overset{\pi_0(r)}{\to} \pi_0(A)$$

has to be the identity. This is impossible.

Definition. (Homotopic paths) The paths $\gamma, \gamma' : I \to X$ are *homotopic as paths* if they are homotopic rel $\{0,1\} \subseteq I$.

Lemma. If $\gamma_0, \gamma_1 : x_0 \rightsquigarrow x_1$ and $\gamma_0', \gamma_1' : x_1 \rightsquigarrow x_2$ are homotopic then $\gamma_0 \circ \gamma_0' \simeq \gamma_1 \circ \gamma_1' : x_0 \rightsquigarrow x_2$.

Proof. Let H be the homotopy of paths from γ_0 to γ_1 and let H' be the homotopy of paths from γ_0' to γ_1'. Define

$$H''(s,t) = \begin{cases} H(2s,t) & 0 \le s \le \frac{1}{2} \\ H'(2s-1,t) & \frac{1}{2} \le s \le 1 \end{cases}$$

which is continuous by the gluing lemma. At $H''(s,0) = (\gamma_0 \circ \gamma_0')(s)$ and $H''(s,1) = \gamma_1 \circ \gamma_1'$. So H'' is a homotopy from $\gamma_0 \circ \gamma_0'$ to $\gamma_1 \circ \gamma_1'$.

Proposition. Let $\gamma_0 : x_0 \rightsquigarrow x_1, \gamma_1 : x_1 \rightsquigarrow x_2, \gamma_2 : x_2 \rightsquigarrow x_3$ be paths in X. Then
(i) $(\gamma_0 \circ \gamma_1) \circ \gamma_2 \simeq \gamma_0 \circ (\gamma_1 \circ \gamma_2)$;
(ii) $\gamma_0 \circ c_{x_1} \simeq \gamma_0 \simeq c_{x_0} \circ \gamma_0$;
(iii) $\gamma_0 \gamma_0^{-1} \simeq c_{x_0}$, $\gamma_0^{-1} \circ \gamma_0 \simeq c_{x_1}$.

Proof. From these, we arrive at the following formulae.

(i) The homotopy

$$H(s,t) = \begin{cases} \gamma_0\left(\dfrac{4s}{t+1}\right) & 0 \le s \le \dfrac{t+1}{4}, \\ \gamma_1(4s-1-t) & \dfrac{t+1}{4} \le s \le \dfrac{t+2}{4}, \\ \gamma_2\left(1 - \dfrac{4(1-s)}{2-t}\right) & \dfrac{t+2}{4} \le s \le 1. \end{cases}$$

goes from $(\gamma_0 \cdot \gamma_1) \cdot \gamma_2$ to $\gamma_0 \cdot (\gamma_1 \cdot \gamma_2)$ as paths from x_0 to x_3.

(ii) The homotopy

$$H(s,t) = \begin{cases} \gamma_0\left(\dfrac{2s}{t+1}\right) & 0 \le s \le \dfrac{t+1}{2}, \\ x_1 & \dfrac{t+1}{2} \le s \le 1, \end{cases}$$

goes from $\gamma_0 \cdot c_{x_1}$ to γ_0 as paths from x_0 to x_1.

(iii) The homotopy

$$H(s,t) = \begin{cases} \gamma_0(2s) & 0 \le s \le \dfrac{1-t}{2}, \\ \gamma_0(1-t) & \dfrac{1-t}{2} \le s \le \dfrac{1+t}{2}, \\ \gamma_0(2-2s) & \dfrac{1+t}{2} \le s \le 1. \end{cases}$$

goes from $\gamma_0 \cdot \overline{\gamma_0}$ to c_{x_0} as paths from x_0 to x_0.

Proposition. Let X be a space, $x_0 \in X$. Define

$$\pi_1(X, x_0) = \{\text{loops in } X \text{ based at } x_0\}/\text{homotopy}.$$

Then the rule $[\gamma] \circ [\gamma'] = [\gamma \circ \gamma']$ defines a group structure on $\pi_1(X, x_0)$. The identity is paths homotopic to the constant path.

Proof. The previous proposition.

Definition. (Fundamental Group) We call $\pi_1(X, x_0)$ the *fundamental group* of X at x_0.